

Supplementary material for “Phase-Dependent False-Alarm Calibration and Its Transport Across Engines in Full-Flight Aero-Engine Anomaly Detection”

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This supplement accompanies the article. Every table is a formatted view of committed outputs of the frozen and post-confirmation analyses; values are rounded for display and nothing is recomputed. Healthy false-positive rates (FPR) are percentages of healthy rows; differences are in percentage points (pp). Runs are one PCA run (PCA) and three seeds each of Isolation Forest (IF s0–s2) and the past-only LSTM (LSTM s0–s2). Arms: P, pooled; C, retrospective phase-conditioned; C', past-only regime-conditioned; Q, continuous contextual baseline (healthy side only). The primary nominal target is 1%. Machine-readable versions of all tables are in the public repository (<https://github.com/surnamemei/ncmapss-phase-entanglement>; [results/](#) and [paper/mssp/evidence/](#)).

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1 Provenance, reproduction and deviations

The study was staged: DS02 exploratory discovery, a protocol freeze, a one-shot frozen DS03 confirmation, and two post-confirmation stages (protocol v1.0 and the adversarial-validation plan), each committed before execution. Before every post-confirmation run, reproduction gates required exact agreement with the frozen evidence: rescored healthy score vectors equalled the frozen fingerprints bit for bit; recomputed healthy count tables, calibration-only κ locks and bootstrap summaries equalled the committed outputs; and the committed per-flight alarm fractions reproduced every committed detection delay. A hash manifest of 186 frozen files was verified before and after every run. Abnormal-state ($hs = 0$) rows were read once per dataset, after the calibration-only lock, and never again. Table S1 lists the governing documents and records.

Deviations and amendments. (i) Amendment A-1 corrected a repository-metadata statement about the published v1.0.0 release tag; it was recorded before any abnormal-state row was opened and changed no analysis. (ii) Dev-1: the error decomposition (Section 4) was added after the plan and is descriptive. (iii) Dev-2: operational definitions of the target-robustness claims were fixed in code before execution. (iv) Dev-3: after all runs, an output guard and a robustness fix for phases without rows were added to the analysis module; no output was regenerated, and every output names the module version that produced it. (v) Story-lock Amendment 1 corrected the count of audit engines from flight classes absent from calibration (four, three of them class 1) without changing any analysis.

Verification. The repository’s test suite re-derives the post-confirmation metrics and operating points from committed inputs, and a checker re-derives every number cited in the article and verifies that it appears verbatim.

Table S1: Provenance of the analyses reported in the article. Each run record also names the analysis-module version that produced its outputs.

Artefact	Path	SHA-256 (first 16 hex)
Frozen baseline manifest (186 files)	docs/mssp/frozen_baseline.sha256	37c0cef26826bef2
Post-confirmation protocol v1.0	docs/mssp/post_confirmation_mitigation_protocol.md	b4a669cdd00d4075
Protocol amendments	docs/mssp/AMENDMENTS.md	60522a5b4eb5be60
Adversarial validation plan	docs/mssp/adversarial_validation_plan.md	d993dd3f9e5ab286
Final story lock	docs/mssp/FINAL_STORY_LOCK.md	f6c5b2cf2d067ed0
Post-confirmation module	src/mssp_mitigation.py	2764f27113a0c7af
Adversarial-validation module	src/mssp_adversarial.py	0b2db2b4ad60e509
DS02 endpoint record (abnormal rows read once)	results/mssp_mitigation/ds02/stage_e/stage_e_record.json	a2ecfaa0eea41074
DS03 endpoint record (abnormal rows read once)	results/mssp_mitigation/ds03/stage_e/stage_e_record.json	e95e80c76d154ec7
DS02 adversarial run record (A–C)	results/mssp_adversarial/ds02/run_record_ABC.json	f676c2189422e2b8
DS03 adversarial run record (A–C)	results/mssp_adversarial/ds03/run_record_ABC.json	5fdd9e77f976964c
DS02 baseline run record (D)	results/mssp_adversarial/ds02/run_record_D.json	df0d765dd5ed67dd
DS03 baseline run record (D)	results/mssp_adversarial/ds03/run_record_D.json	1c48684e825e8c9c
Adversarial summary record	results/mssp_adversarial/summary/summary_record.json	1a59f7170292eead

2 Frozen discovery and confirmation

The values in this section are the frozen DS02 (exploratory) and DS03 (frozen confirmation) results, shown unchanged.

Table S2: Frozen DS02 PCA correction sensitivity (exploratory): pooled healthy FPR (%) by phase at the 1% target, primary phase rule, and worst off-diagonal cross-phase transfer FPR (%).

Correction	Overall	Climb	Cruise	Descent	Worst transfer
static	1.277	0.146	0.583	3.001	14.306
static-derivative	1.239	0.061	0.457	3.106	15.865
history	1.363	0.125	0.428	3.459	15.555

Table S3: Frozen DS02 alternative retrospective phase rule (exploratory; 85% altitude range and centred 60-s altitude rate at most 2 ft/s): pooled healthy FPR (%) at the 1% target.

Run	Climb	Cruise	Descent	Descent – climb (pp)	Descent – cruise (pp)
PCA	0.131	0.270	3.563	+3.432	+3.293

(continued)

Run	Climb	Cruise	Descent	Descent – climb (pp)	Descent – cruise (pp)
IF s0	0.276	0.379	2.515	+2.238	+2.135
IF s1	0.247	0.387	2.787	+2.540	+2.400
IF s2	0.259	0.396	2.629	+2.370	+2.233
LSTM s0	0.451	0.516	2.264	+1.813	+1.747
LSTM s1	0.459	0.524	2.316	+1.857	+1.792
LSTM s2	0.530	0.540	2.270	+1.741	+1.730

Table S4: Frozen individual-run pooled healthy FPR (%) at the 1% target with hierarchical (engine, then flight) bootstrap 95% intervals for the contrasts (2000 replicates, thresholds re-estimated). Intervals belong to individual runs; no seed-mean interval exists.

Dataset	Run	Climb	Cruise	Descent	Descent – climb, 95% CI (pp)	Descent – cruise, 95% CI (pp)
DS02	PCA	0.125	0.428	3.459	+3.335 [+2.070, +5.629]	+3.031 [+1.808, +5.261]
	IF s0	0.285	0.566	2.335	+2.050 [+0.850, +4.074]	+1.769 [+0.678, +3.749]
	IF s1	0.253	0.528	2.669	+2.416 [+1.313, +4.166]	+2.141 [+1.012, +3.943]
	IF s2	0.266	0.580	2.455	+2.190 [+1.063, +3.947]	+1.875 [+0.742, +3.709]
	LSTM s0	0.452	0.707	2.078	+1.626 [+0.317, +2.838]	+1.371 [+0.528, +2.490]
	LSTM s1	0.474	0.731	2.102	+1.628 [−0.022, +2.817]	+1.372 [+0.399, +2.454]
	LSTM s2	0.539	0.750	2.060	+1.521 [−0.347, +2.802]	+1.309 [+0.330, +2.511]
DS03	PCA	0.877	0.912	1.977	+1.100 [+0.434, +1.785]	+1.066 [+0.560, +1.573]
	IF s0	0.417	1.095	2.166	+1.749 [+0.901, +2.847]	+1.071 [+0.404, +1.994]
	IF s1	0.399	1.060	2.127	+1.728 [+0.891, +2.833]	+1.067 [+0.426, +1.976]
	IF s2	0.412	1.157	2.134	+1.722 [+0.871, +2.772]	+0.976 [+0.347, +1.862]
	LSTM s0	0.759	1.795	2.231	+1.472 [+0.536, +2.870]	+0.436 [−0.222, +1.201]
	LSTM s1	0.719	1.890	2.220	+1.501 [+0.512, +3.022]	+0.330 [−0.270, +1.158]
	LSTM s2	0.774	1.982	2.193	+1.419 [+0.379, +3.059]	+0.210 [−0.582, +1.025]

Table S5: Frozen per-engine pooled-threshold healthy FPR (%) at the 1% target (DS02 exploratory; DS03 frozen confirmation). Directional exceptions remain visible.

Dataset	Run	Engine	Climb	Cruise	Descent	Descent – climb (pp)	Descent – cruise (pp)
DS02	PCA	11	0.097	0.406	2.780	+2.683	+2.374
		14	0.294	0.491	4.972	+4.678	+4.481
		15	0.113	0.425	3.666	+3.553	+3.241
	IF s0	11	0.091	0.371	2.234	+2.143	+1.863
		14	1.487	1.351	2.510	+1.024	+1.159
		15	0.197	0.393	2.396	+2.199	+2.003
	IF s1	11	0.082	0.304	2.348	+2.266	+2.044
		14	1.291	1.393	3.287	+1.996	+1.894
		15	0.180	0.352	2.824	+2.644	+2.473
	IF s2	11	0.078	0.373	2.298	+2.220	+1.925
		14	1.389	1.432	2.724	+1.335	+1.292
		15	0.191	0.383	2.551	+2.360	+2.167
	LSTM s0	11	0.227	0.558	2.364	+2.137	+1.806
		14	1.887	1.282	2.071	+0.184	+0.789
		15	0.340	0.589	1.621	+1.281	+1.032
	LSTM s1	11	0.246	0.615	2.523	+2.277	+1.908
		14	1.993	1.264	1.835	−0.158	+0.571
		15	0.344	0.585	1.581	+1.237	+0.996
	LSTM s2	11	0.248	0.592	2.378	+2.130	+1.786
		14	2.296	1.384	1.775	−0.521	+0.391
		15	0.418	0.612	1.714	+1.296	+1.102
DS03	PCA	10	1.258	1.247	2.448	+1.190	+1.201
		11	0.383	0.878	2.173	+1.790	+1.294

(continued)

Dataset	Run	Engine	Climb	Cruise	Descent	Descent – climb (pp)	Descent – cruise (pp)
IF s0		12	0.322	1.039	1.061	+0.739	+0.022
		13	1.120	0.744	2.249	+1.129	+1.505
		14	0.744	0.587	1.431	+0.686	+0.843
		15	1.112	0.846	2.153	+1.041	+1.307
		10	0.374	1.197	3.113	+2.739	+1.917
		11	0.288	1.158	2.281	+1.993	+1.123
		12	0.847	0.617	1.845	+0.998	+1.229
		13	0.453	1.274	2.665	+2.212	+1.390
		14	0.288	1.095	1.282	+0.994	+0.187
		15	0.378	0.971	1.495	+1.117	+0.524
	IF s1	10	0.335	1.171	3.155	+2.820	+1.984
		11	0.307	1.117	2.318	+2.011	+1.201
		12	0.839	0.550	1.696	+0.857	+1.146
		13	0.419	1.378	2.604	+2.185	+1.226
		14	0.276	0.910	1.105	+0.829	+0.196
IF s2		15	0.355	0.901	1.516	+1.161	+0.615
		10	0.389	1.314	3.232	+2.844	+1.918
		11	0.296	1.213	2.290	+1.994	+1.077
		12	0.808	0.573	1.689	+0.881	+1.116
		13	0.450	1.428	2.619	+2.169	+1.190
		14	0.258	1.089	1.169	+0.911	+0.080
		15	0.372	0.981	1.446	+1.074	+0.466
	LSTM s0	10	0.703	2.519	3.696	+2.992	+1.176
		11	0.545	1.956	2.671	+2.126	+0.714
		12	1.141	0.585	1.237	+0.097	+0.652
		13	0.913	2.397	2.690	+1.777	+0.292
		14	0.693	0.904	0.992	+0.299	+0.088
LSTM s1		15	0.693	1.323	1.556	+0.863	+0.233
		10	0.705	2.793	3.906	+3.201	+1.113
		11	0.504	2.041	2.619	+2.115	+0.579
		12	1.093	0.614	1.210	+0.117	+0.595
		13	0.842	2.412	2.667	+1.825	+0.254
		14	0.681	0.849	0.918	+0.237	+0.070
		15	0.634	1.445	1.449	+0.815	+0.004
	LSTM s2	10	0.713	2.934	4.083	+3.370	+1.149
		11	0.550	2.141	2.652	+2.102	+0.510
		12	1.036	0.487	0.950	−0.086	+0.463
		13	0.951	2.810	2.705	+1.754	−0.105
		14	0.651	0.746	0.784	+0.133	+0.039
		15	0.792	1.378	1.344	+0.552	−0.034

Table S6: Frozen pooled cross-phase threshold-transfer matrices at the 1% target: healthy FPR (%) when the threshold calibrated within one phase is applied to each phase. Diagonal cells are the within-phase (phase-conditioned) FPRs.

Dataset	Run	Calibrated in	Applied to climb	Applied to cruise	Applied to descent
DS02	PCA	climb	1.403	3.430	15.555
		cruise	0.581	1.327	9.120
		descent	0.041	0.238	1.081
	IF s0	climb	1.655	3.357	29.982
		cruise	1.079	1.845	17.042
		descent	0.108	0.205	0.921
	IF s1	climb	1.544	3.000	25.461
		cruise	0.993	1.528	14.017
		descent	0.103	0.237	0.978
	IF s2	climb	1.686	2.914	26.400
		cruise	1.141	1.715	16.783
		descent	0.110	0.204	0.927
	LSTM s0	climb	1.385	2.371	5.505
		cruise	0.698	1.115	3.039
		descent	0.237	0.280	1.080
	LSTM s1	climb	1.243	2.437	6.774

(continued)

Dataset	Run	Calibrated in	Applied to climb	Applied to cruise	Applied to descent
DS03	LSTM s2	cruise	0.729	1.279	3.370
		descent	0.235	0.224	0.961
		climb	1.456	2.391	5.582
		cruise	0.766	1.073	2.953
		descent	0.251	0.269	1.001
	PCA	climb	1.856	3.281	6.716
		cruise	1.259	1.789	3.659
		descent	0.584	0.578	1.231
	IF s0	climb	1.656	3.294	9.194
		cruise	0.531	1.394	2.930
		descent	0.322	0.664	1.271
	IF s1	climb	1.663	3.389	9.438
		cruise	0.485	1.366	2.881
		descent	0.306	0.634	1.267
	IF s2	climb	1.745	3.666	9.525
		cruise	0.503	1.424	2.689
		descent	0.313	0.672	1.268
	LSTM s0	climb	1.777	3.978	5.112
		cruise	0.773	1.830	2.270
		descent	0.511	1.086	1.488
	LSTM s1	climb	1.845	4.313	5.617
		cruise	0.669	1.756	2.059
		descent	0.484	1.205	1.495
	LSTM s2	climb	1.796	4.193	5.195
		cruise	0.671	1.720	1.887
		descent	0.545	1.338	1.529

Table S7: Frozen flight-level burden under pooled calibration at the 1% target: percentage of healthy audit flights with any alarm in each phase, and alarm events per healthy flight. This endpoint is distinct from the κ -based healthy-flight false-flag rate of the post-confirmation analyses.

Dataset	Run	Any alarm, climb (%)	Any alarm, cruise (%)	Any alarm, descent (%)	Events per flight, climb	Events per flight, cruise	Events per flight, descent
DS02	PCA	13.2	50.0	98.7	0.32	4.43	16.03
	IF s0	68.4	44.7	86.8	1.03	3.07	9.08
	IF s1	60.5	44.7	85.5	0.88	2.76	8.36
	IF s2	67.1	46.1	86.8	0.97	3.07	9.61
	LSTM s0	97.4	77.6	100.0	1.76	4.66	8.62
	LSTM s1	92.1	64.5	100.0	1.45	4.12	9.68
	LSTM s2	98.7	68.4	100.0	1.75	4.49	8.86
DS03	PCA	67.6	73.6	99.3	4.57	10.40	30.66
	IF s0	81.1	67.6	99.3	2.93	6.76	22.95
	IF s1	81.1	65.5	99.3	2.59	6.44	20.99
	IF s2	77.7	66.2	99.3	2.81	7.07	21.39
	LSTM s0	100.0	95.3	100.0	5.96	13.47	26.52
	LSTM s1	100.0	92.6	100.0	5.68	13.28	23.97
	LSTM s2	100.0	94.6	99.3	5.39	13.12	23.39

LSTM training and convergence

Figures S1 and S2 reproduce the recorded training and epoch-selection validation losses without smoothing. The DS02 seed-0 selection reached the pre-specified 600-epoch ceiling while validation loss was still improving; it is reported as a convergence limitation and was not retuned. These diagnostics do not inspect test outcomes.

3 Post-confirmation protocol v1.0

These analyses are exploratory. They reuse the frozen detectors and engine partitions and are not part of the DS03 confirmation.

DS02 exploratory: frozen LSTM epoch-selection histories

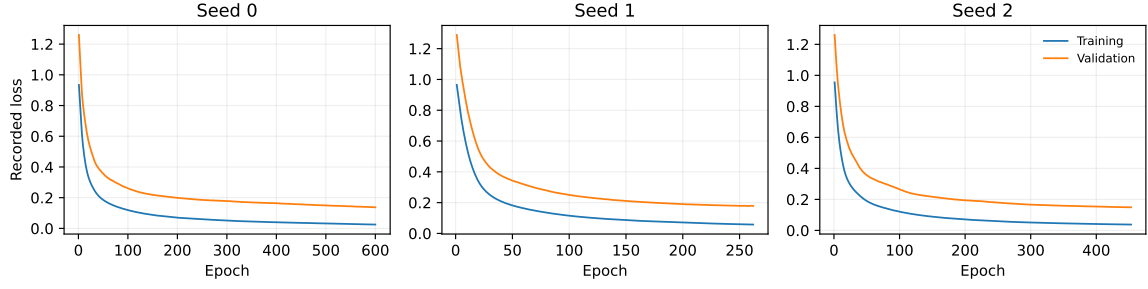


Figure S1: Frozen DS02 LSTM training and epoch-selection validation losses (seeds 0–2).

DS03 confirmatory: frozen LSTM epoch-selection histories

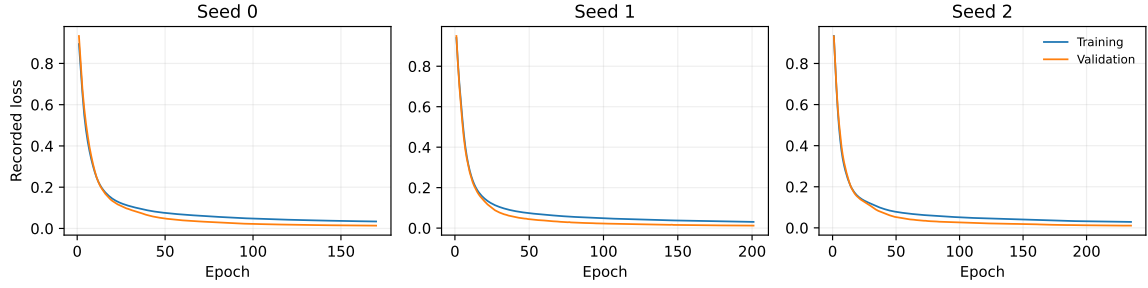


Figure S2: Frozen DS03 LSTM training and epoch-selection validation losses (seeds 0–2).

Table S8: Post-confirmation pooled audit healthy FPR (%) by phase at the 1% target under P, C and C' (protocol v1.0).

Dataset	Run	Arm	Climb	Cruise	Descent	Overall
DS02	PCA	P	0.125	0.428	3.459	1.363
		C	1.403	1.327	1.081	1.265
		C'	1.462	1.075	1.360	1.273
	IF s0	P	0.285	0.566	2.335	1.084
		C	1.655	1.845	0.921	1.485
		C'	1.525	1.119	1.669	1.410
	IF s1	P	0.253	0.528	2.669	1.172
		C	1.544	1.528	0.978	1.348
		C'	1.434	1.099	1.699	1.389
	IF s2	P	0.266	0.580	2.455	1.125
		C	1.686	1.715	0.927	1.443
		C'	1.560	1.125	1.680	1.426
	LSTM s0	P	0.452	0.707	2.078	1.099
		C	1.385	1.115	1.080	1.175
		C'	1.377	1.541	1.778	1.577
	LSTM s1	P	0.474	0.731	2.102	1.122
		C	1.243	1.279	0.961	1.163
		C'	1.200	1.390	1.725	1.452
	LSTM s2	P	0.539	0.750	2.060	1.133
		C	1.456	1.073	1.001	1.150
		C'	1.419	1.503	1.748	1.563
DS03	PCA	P	0.877	0.912	1.977	1.341
		C	1.856	1.789	1.231	1.577
		C'	1.977	1.108	1.685	1.564
	IF s0	P	0.417	1.095	2.166	1.364
		C	1.656	1.394	1.271	1.410
		C'	1.053	0.827	1.694	1.240
	IF s1	P	0.399	1.060	2.127	1.332
		C	1.663	1.366	1.267	1.400
		C'	1.037	0.791	1.672	1.215
	IF s2	P	0.412	1.157	2.134	1.370

(continued)

Dataset	Run	Arm	Climb	Cruise	Descent	Overall
		C	1.745	1.424	1.268	1.441
		C'	1.083	0.882	1.719	1.276
	LSTM s0	P	0.759	1.795	2.231	1.713
		C	1.777	1.830	1.488	1.676
		C'	1.841	1.833	1.723	1.790
	LSTM s1	P	0.719	1.890	2.220	1.730
		C	1.845	1.756	1.495	1.671
		C'	1.879	1.931	1.625	1.792
	LSTM s2	P	0.774	1.982	2.193	1.764
		C	1.796	1.720	1.529	1.661
		C'	1.877	2.004	1.746	1.866

Table S9: Paired arm-minus-P bootstrap 95% intervals (percentage points) at the 1% target for the row-pooled calibration metrics (frozen U1 resampling plans; engines then flights; thresholds re-estimated; 2000 replicates). A1, A2, A3 and A5 are row-pooled; the last column weights audit engines equally. Every interval includes zero.

Dataset	Run	Arm - P	A1 (pp)	A2 (pp)	A3 (pp)	A5 (pp)	A2, equal-engine (pp)
DS02	PCA	C	[-4.28, +0.63]	[-3.23, +1.21]	[-1.82, +0.85]	[-0.52, +1.21]	[-3.24, +2.90]
		C'	[-4.31, +1.49]	[-3.39, +2.83]	[-1.94, +1.77]	[-0.59, +1.74]	[-3.30, +5.81]
	IF s0	C	[-3.02, +4.92]	[-2.26, +4.34]	[-1.23, +3.03]	[-0.27, +1.86]	[-2.11, +5.41]
		C'	[-2.93, +3.06]	[-2.25, +3.92]	[-1.25, +2.80]	[-0.37, +1.99]	[-2.09, +5.04]
	IF s1	C	[-3.37, +3.92]	[-2.54, +3.23]	[-1.40, +2.36]	[-0.31, +1.26]	[-2.35, +4.52]
		C'	[-3.22, +2.06]	[-2.52, +3.22]	[-1.43, +2.35]	[-0.34, +1.69]	[-2.22, +4.24]
	IF s2	C	[-3.02, +5.30]	[-2.21, +4.51]	[-1.26, +3.25]	[-0.26, +1.77]	[-2.17, +5.94]
		C'	[-2.96, +3.31]	[-2.18, +4.28]	[-1.25, +2.89]	[-0.32, +1.97]	[-2.01, +5.69]
	LSTM s0	C	[-2.01, +3.89]	[-1.40, +3.65]	[-0.82, +2.13]	[-0.21, +1.25]	[-1.28, +5.65]
		C'	[-1.88, +2.31]	[-1.13, +3.38]	[-0.66, +2.47]	[-0.23, +2.20]	[-1.16, +5.32]
	LSTM s1	C	[-2.11, +3.18]	[-1.51, +2.66]	[-0.89, +1.66]	[-0.24, +1.03]	[-1.39, +3.37]
		C'	[-1.74, +1.86]	[-1.06, +2.49]	[-0.63, +2.09]	[-0.22, +1.86]	[-1.04, +3.11]
	LSTM s2	C	[-2.06, +4.03]	[-1.51, +3.67]	[-0.87, +2.16]	[-0.24, +1.01]	[-1.42, +5.95]
		C'	[-1.89, +2.62]	[-1.29, +3.35]	[-0.72, +2.56]	[-0.25, +2.17]	[-1.27, +5.51]
DS03	PCA	C	[-1.13, +2.56]	[-0.56, +2.61]	[-0.27, +1.51]	[-0.05, +0.92]	[-0.55, +2.43]
		C'	[-0.97, +2.91]	[-0.50, +3.67]	[-0.26, +2.21]	[-0.15, +1.48]	[-0.49, +3.61]
	IF s0	C	[-1.99, +4.43]	[-1.08, +4.85]	[-0.56, +2.68]	[-0.25, +1.34]	[-1.08, +5.00]
		C'	[-1.49, +1.82]	[-0.91, +2.49]	[-0.52, +1.39]	[-0.39, +0.99]	[-0.90, +2.51]
	IF s1	C	[-1.94, +4.25]	[-1.09, +4.70]	[-0.56, +2.63]	[-0.24, +1.35]	[-1.09, +4.69]
		C'	[-1.51, +1.71]	[-0.89, +2.46]	[-0.52, +1.40]	[-0.40, +1.01]	[-0.87, +2.42]
	IF s2	C	[-1.90, +4.88]	[-0.99, +5.27]	[-0.51, +2.96]	[-0.20, +1.44]	[-0.99, +5.47]
		C'	[-1.52, +2.12]	[-0.84, +2.85]	[-0.50, +1.65]	[-0.36, +1.11]	[-0.84, +2.84]
	LSTM s0	C	[-1.73, +1.82]	[-0.82, +1.97]	[-0.37, +1.08]	[-0.23, +0.45]	[-0.88, +2.45]
		C'	[-1.90, +1.08]	[-1.09, +1.65]	[-0.64, +0.97]	[-0.48, +0.77]	[-1.08, +2.04]
	LSTM s1	C	[-1.83, +1.84]	[-0.83, +1.89]	[-0.40, +1.01]	[-0.24, +0.43]	[-0.89, +2.35]
		C'	[-2.03, +1.11]	[-1.08, +1.52]	[-0.64, +0.91]	[-0.48, +0.68]	[-1.10, +1.92]
	LSTM s2	C	[-1.69, +2.52]	[-0.81, +2.48]	[-0.43, +1.38]	[-0.29, +0.59]	[-0.85, +3.05]
		C'	[-1.89, +1.57]	[-1.08, +2.07]	[-0.70, +1.18]	[-0.56, +0.92]	[-1.07, +2.49]

Table S10: Target robustness of the post-confirmation claims: number of runs (of 7) in which each claim holds at the nominal targets 0.5%, 1% and 2%, with the pre-specified classification.

Dataset	Claim	0.5%	1%	2%	Class
DS02	R1 descent is the highest pooled healthy-FPR phase (P)	7/7	7/7	7/7	invariant
	R2 C reduces between-phase disparity (A1)	7/7	7/7	7/7	invariant
	R3 C reduces max-abs nominal calibration error (A2)	7/7	7/7	7/7	invariant
	R4 C reduces RMS nominal calibration error (A3)	7/7	7/7	7/7	invariant
	R5 C mainly redistributes: overall FPR change < 0.5 × A1 change	6/7	7/7	7/7	mostly invariant
	R6 C abnormal-state row alarm-rate ratio ≥ 0.9	7/7	7/7	7/7	invariant
	R8 C is not earlier than P at matched false-flag burden	4/7	4/7	3/7	reversed
	R2 C' reduces between-phase disparity (A1)	7/7	7/7	7/7	invariant
	R3 C' reduces max-abs nominal calibration error (A2)	4/7	7/7	7/7	mostly invariant
	R4 C' reduces RMS nominal calibration error (A3)	4/7	7/7	7/7	mostly invariant
	R5 C' mainly redistributes: overall FPR change < 0.5 × A1 change	2/7	7/7	7/7	reversed
	R6 C' abnormal-state row alarm-rate ratio ≥ 0.9	7/7	7/7	7/7	invariant

(continued)

Dataset	Claim	0.5%	1%	2%	Class
	R8 C' is not earlier than P at matched false-flag burden	7/7	3/7	2/7	not supported at 1%
	R7 realized pooled healthy-flight FFR exceeds nominal 5% (P)	7/7	7/7	7/7	invariant
	R7 realized pooled healthy-flight FFR exceeds nominal 5% (C)	7/7	7/7	7/7	invariant
	R7 realized pooled healthy-flight FFR exceeds nominal 5% (C')	7/7	7/7	7/7	invariant
DS03	R1 descent is the highest pooled healthy-FPR phase (P)	7/7	7/7	7/7	invariant
	R2 C reduces between-phase disparity (A1)	7/7	7/7	7/7	invariant
	R3 C reduces max-abs nominal calibration error (A2)	7/7	7/7	6/7	mostly invariant
	R4 C reduces RMS nominal calibration error (A3)	7/7	6/7	6/7	mostly invariant
	R5 C mainly redistributes: $ \text{overall FPR change} < 0.5 \times A1 \text{ change} $	7/7	7/7	7/7	invariant
	R6 C abnormal-state row alarm-rate ratio ≥ 0.9	7/7	7/7	7/7	invariant
	R8 C is not earlier than P at matched false-flag burden	6/7	7/7	6/7	mostly invariant
	R2 C' reduces between-phase disparity (A1)	7/7	7/7	7/7	invariant
	R3 C' reduces max-abs nominal calibration error (A2)	6/7	7/7	6/7	mostly invariant
	R4 C' reduces RMS nominal calibration error (A3)	4/7	6/7	5/7	target-sensitive
	R5 C' mainly redistributes: $ \text{overall FPR change} < 0.5 \times A1 \text{ change} $	6/7	6/7	6/7	mostly invariant
	R6 C' abnormal-state row alarm-rate ratio ≥ 0.9	7/7	7/7	7/7	invariant
	R8 C' is not earlier than P at matched false-flag burden	6/7	5/7	7/7	mostly invariant
	R7 realized pooled healthy-flight FFR exceeds nominal 5% (P)	7/7	6/7	7/7	mostly invariant
	R7 realized pooled healthy-flight FFR exceeds nominal 5% (C)	4/7	4/7	7/7	target-sensitive
	R7 realized pooled healthy-flight FFR exceeds nominal 5% (C')	7/7	7/7	7/7	invariant

4 Adversarial validation

Table S11: Per-engine maximum nominal calibration error A2 (percentage points) at the 1% target for every detector run under P, C, C' and Q.

Dataset	Engine (class)	Run	P	C	C'	Q
DS02	11 (3)	PCA	1.78	0.38	0.63	2.00
		IF s0	1.23	0.25	0.97	1.45
		IF s1	1.35	0.35	0.99	1.28
		IF s2	1.30	0.25	0.97	1.48
		LSTM s0	1.36	0.36	1.11	0.61
		LSTM s1	1.52	0.38	1.15	0.44
		LSTM s2	1.38	0.36	1.07	4.25
	14 (1)	PCA	3.97	4.31	5.31	2.20
		IF s0	1.51	6.42	6.12	2.16
		IF s1	2.29	5.96	5.72	1.58
		IF s2	1.72	6.59	6.27	1.95
		LSTM s0	1.07	4.42	4.39	1.30
		LSTM s1	0.99	3.74	3.67	0.54
		LSTM s2	1.30	4.65	4.50	1.39
	15 (2)	PCA	2.67	0.28	0.40	0.58
		IF s0	1.40	0.27	0.57	3.63
		IF s1	1.82	0.26	0.56	1.08
		IF s2	1.55	0.28	0.55	4.15
		LSTM s0	0.66	0.19	0.23	0.62
		LSTM s1	0.66	0.14	0.29	0.64
		LSTM s2	0.71	0.30	0.27	4.18
DS03	10 (3)	PCA	1.45	1.52	1.77	7.57
		IF s0	2.11	2.37	1.30	6.31
		IF s1	2.16	2.06	1.36	6.27
		IF s2	2.23	2.56	1.33	6.44
		LSTM s0	2.70	1.56	1.93	3.67
		LSTM s1	2.91	1.57	2.21	4.91
		LSTM s2	3.08	1.69	2.39	3.59
	11 (3)	PCA	1.17	0.97	1.14	0.56
		IF s0	1.28	0.52	0.93	0.89
		IF s1	1.32	0.49	0.93	0.87
		IF s2	1.29	0.50	0.96	0.89
		LSTM s0	1.67	0.99	1.16	0.84
		LSTM s1	1.62	0.91	1.02	0.81
		LSTM s2	1.65	0.97	1.16	0.76
	12 (1)	PCA	0.68	0.82	0.73	0.69
		IF s0	0.85	1.07	0.75	0.40
		IF s1	0.70	0.92	0.70	0.49
		IF s2	0.69	0.96	0.77	0.50

(continued)

Dataset	Engine (class)	Run	P	C	C'	Q
13 (3)		LSTM s0	0.41	2.04	2.24	0.72
		LSTM s1	0.39	2.17	2.47	0.70
		LSTM s2	0.51	2.02	2.39	0.74
		PCA	1.25	1.22	1.42	1.41
		IF s0	1.66	0.77	0.77	1.69
		IF s1	1.60	0.86	0.75	1.24
		IF s2	1.62	0.81	0.80	1.33
		LSTM s0	1.69	1.47	1.10	0.64
		LSTM s1	1.67	1.22	1.20	0.70
		LSTM s2	1.81	1.39	1.21	0.58
		PCA	0.43	0.37	0.43	0.30
		IF s0	0.71	0.44	0.50	0.94
14 (1)		IF s1	0.72	0.44	0.44	0.95
		IF s2	0.74	0.47	0.45	0.97
		LSTM s0	0.31	0.64	0.74	0.73
		LSTM s1	0.32	0.76	0.84	0.66
		LSTM s2	0.35	0.63	0.79	0.78
		PCA	1.15	1.28	1.20	0.54
		IF s0	0.62	0.25	0.34	0.65
		IF s1	0.64	0.40	0.35	0.62
		IF s2	0.63	0.38	0.43	0.66
		LSTM s0	0.56	0.65	0.69	0.50
		LSTM s1	0.45	0.54	0.54	0.49
		LSTM s2	0.38	0.74	0.83	0.61

Table S12: Denominator audit: sign agreement between row-pooled and alternative weightings or endpoints, over runs, targets and arms. The frozen burden endpoint (healthy flights with any alarm) and the κ -based healthy-flight false-flag rate are distinct quantities.

Dataset	Check	Agreeing
DS02	row-pooled vs equal-engine sign of diff_spread	25 of 42
	row-pooled vs equal-engine sign of diff_max_abs_error	16 of 42
	row-pooled vs equal-engine sign of diff_rms_error	15 of 42
	equal-flight vs row-pooled sign of overall healthy FPR change (phase_conditioned)	19 of 21
	equal-flight vs row-pooled sign of overall healthy FPR change (causal_regime)	19 of 21
	‘healthy flights with any alarm’ vs kappa FFR sign of change (phase_conditioned)	8 of 21
	‘healthy flights with any alarm’ vs kappa FFR sign of change (causal_regime)	8 of 21
	row abnormal alarm rate higher vs flight delay earlier (phase_conditioned)	5 of 21
	row abnormal alarm rate higher vs flight delay earlier (causal_regime)	13 of 21
DS03	row-pooled vs equal-engine sign of diff_spread	42 of 42
	row-pooled vs equal-engine sign of diff_max_abs_error	39 of 42
	row-pooled vs equal-engine sign of diff_rms_error	36 of 42
	equal-flight vs row-pooled sign of overall healthy FPR change (phase_conditioned)	14 of 21
	equal-flight vs row-pooled sign of overall healthy FPR change (causal_regime)	19 of 21
	‘healthy flights with any alarm’ vs kappa FFR sign of change (phase_conditioned)	0 of 21
	‘healthy flights with any alarm’ vs kappa FFR sign of change (causal_regime)	4 of 21
	row abnormal alarm rate higher vs flight delay earlier (phase_conditioned)	10 of 21
	row abnormal alarm rate higher vs flight delay earlier (causal_regime)	13 of 21

Matched false-flag delay

Figure S3 shows the full operating characteristics: the lower-median detection delay against the realized pooled healthy-flight false-flag rate as κ is swept over the calibration-only grid. Tables S13 and S14 give the matched comparisons at the pre-specified anchors, and Table S15 gives per-engine delays at the locked κ together with their false-flag rates.

Table S13: Matched healthy-flight false-flag comparisons at the 1% row target (nearest-anchor rule): realized pooled false-flag rates, lower-median delays (flights from the labelled abnormal-state onset), per-engine earlier/same/late counts, and the paired lower-median difference (arm – P). Unsupported anchors are shown but were not used.

Dataset	Run	Arm	Anchor (%)	Supported	FFR (%) P / arm	Median delay P / arm	Engines earlier/ same/late	Paired median diff.
DS02	PCA	C	2.5	yes	2.6 / 2.6	19 / 2	3/0/0	–13
		C	5	yes	5.3 / 5.3	19 / 2	3/0/0	–13

(continued)

Dataset	Run	Arm	Anchor (%)	Supported	FFR (%) P / arm	Median delay P / arm	Engines earlier/ same/later	Paired median diff.
		C	10	no	10.5 / 11.8	4 / 2	1/1/1	0
		C	15	yes	14.5 / 14.5	2 / 2	1/1/1	0
		C	20	yes	19.7 / 19.7	2 / 2	1/1/1	0
		C'	2.5	yes	2.6 / 1.3	19 / 2	3/0/0	-13
		C'	5	yes	5.3 / 5.3	19 / 2	3/0/0	-13
		C'	10	yes	10.5 / 10.5	4 / 2	1/1/1	0
		C'	15	yes	14.5 / 14.5	2 / 1	2/0/1	-2
		C'	20	no	19.7 / 18.4	2 / 0	2/0/1	-2
		C	2.5	yes	2.6 / 1.3	12 / 5	3/0/0	-7
		C	5	yes	5.3 / 5.3	11 / 5	3/0/0	-6
		C	10	yes	10.5 / 10.5	4 / 3	2/1/0	-3
		C	15	yes	15.8 / 14.5	2 / 1	1/2/0	0
		C	20	no	18.4 / 19.7	2 / 1	1/0/2	1
		C'	2.5	yes	2.6 / 1.3	12 / 5	3/0/0	-7
		C'	5	yes	5.3 / 5.3	11 / 5	3/0/0	-6
		C'	10	yes	10.5 / 10.5	4 / 3	2/1/0	-3
		C'	15	yes	15.8 / 14.5	2 / 0	2/1/0	-1
		C'	20	no	18.4 / 19.7	2 / 0	1/1/1	0
		C	2.5	yes	1.3 / 1.3	15 / 5	3/0/0	-7
		C	5	yes	5.3 / 5.3	12 / 5	3/0/0	-7
		C	10	yes	10.5 / 10.5	6 / 3	2/1/0	-5
		C	15	yes	14.5 / 14.5	3 / 1	2/1/0	-2
		C	20	yes	19.7 / 19.7	3 / 1	2/1/0	-2
		C'	2.5	yes	1.3 / 1.3	15 / 5	3/0/0	-8
		C'	5	yes	5.3 / 5.3	12 / 5	3/0/0	-7
		C'	10	yes	10.5 / 10.5	6 / 3	2/1/0	-5
		C'	15	yes	14.5 / 15.8	3 / 0	2/1/0	-2
		C'	20	yes	19.7 / 19.7	3 / 0	2/1/0	-2
		C	2.5	yes	2.6 / 1.3	13 / 5	3/0/0	-7
		C	5	yes	5.3 / 5.3	12 / 4	3/0/0	-7
		C	10	yes	10.5 / 10.5	3 / 3	1/1/1	0
		C	15	yes	14.5 / 14.5	2 / 1	2/0/1	-1
		C	20	yes	19.7 / 19.7	0 / 1	1/1/1	0
		C'	2.5	yes	2.6 / 1.3	13 / 5	3/0/0	-7
		C'	5	yes	5.3 / 5.3	12 / 5	3/0/0	-7
		C'	10	yes	10.5 / 10.5	3 / 1	2/1/0	-2
		C'	15	yes	14.5 / 14.5	2 / 0	2/0/1	-2
		C'	20	yes	19.7 / 19.7	0 / 0	0/2/1	0
		C	2.5	no	3.9 / 1.3	14 / 10	3/0/0	-4
		C	5	yes	5.3 / 5.3	13 / 5	3/0/0	-8
		C	10	no	11.8 / 10.5	11 / 2	2/0/1	-4
		C	15	yes	14.5 / 14.5	2 / 2	0/2/1	0
		C	20	yes	19.7 / 21.1	2 / 2	0/2/1	0
		C'	2.5	no	3.9 / 1.3	14 / 11	3/0/0	-3
		C'	5	yes	5.3 / 5.3	13 / 7	3/0/0	-6
		C'	10	no	11.8 / 10.5	11 / 5	2/0/1	-3
		C'	15	yes	14.5 / 14.5	2 / 5	0/1/2	3
		C'	20	yes	19.7 / 19.7	2 / 5	0/1/2	3
		C	2.5	yes	1.3 / 2.6	13 / 10	3/0/0	-6
		C	5	yes	5.3 / 5.3	13 / 7	3/0/0	-5
		C	10	yes	10.5 / 10.5	2 / 5	1/0/2	1
		C	15	yes	14.5 / 14.5	2 / 2	0/1/2	1
		C	20	yes	19.7 / 21.1	2 / 2	0/2/1	0
		C'	2.5	yes	1.3 / 1.3	13 / 10	3/0/0	-4
		C'	5	yes	5.3 / 5.3	13 / 7	3/0/0	-6
		C'	10	yes	10.5 / 10.5	2 / 7	1/0/2	1
		C'	15	yes	14.5 / 14.5	2 / 7	0/1/2	5
		C'	20	yes	19.7 / 19.7	2 / 7	0/1/2	5
		C	2.5	no	3.9 / 2.6	13 / 10	2/1/0	-4
		C	5	yes	5.3 / 5.3	11 / 2	3/0/0	-8
		C	10	no	7.9 / 9.2	2 / 2	1/1/1	0
		C	15	yes	14.5 / 14.5	2 / 2	0/2/1	0
		C	20	yes	21.1 / 21.1	2 / 2	0/2/1	0
		C'	2.5	no	3.9 / 2.6	13 / 12	3/0/0	-2
		C'	5	yes	5.3 / 3.9	11 / 12	1/1/1	0
		C'	10	no	7.9 / 10.5	2 / 5	1/0/2	1
		C'	15	yes	14.5 / 14.5	2 / 5	0/1/2	3
		C'	20	yes	21.1 / 19.7	2 / 2	0/2/1	0
DS03	PCA	C	2.5	yes	2.7 / 2.7	14 / 12	3/0/3	-4
		C	5	no	4.1 / 4.7	2 / 10	3/0/3	-5
		C	10	yes	10.1 / 10.1	2 / 4	1/2/3	0
		C	15	no	14.2 / 15.5	2 / 2	1/4/1	0

(continued)

Dataset	Run	Arm	Anchor (%)	Supported	FFR (%) P / arm	Median delay P / arm	Engines earlier/ same/later	Paired median diff.
	IF s0	C	20	yes	19.6 / 20.3	2 / 2	1/3/2	0
		C'	2.5	yes	2.7 / 2.7	14 / 10	3/0/3	-4
		C'	5	no	4.1 / 4.7	2 / 10	3/0/3	-4
		C'	10	yes	10.1 / 10.1	2 / 5	1/0/5	1
		C'	15	no	14.2 / 15.5	2 / 5	2/0/4	1
		C'	20	yes	19.6 / 19.6	2 / 1	2/1/3	0
		C	2.5	yes	2.7 / 2.0	6 / 7	0/2/4	1
		C	5	yes	4.7 / 4.7	5 / 7	0/2/4	1
		C	10	yes	10.1 / 10.1	3 / 2	2/2/2	0
		C	15	yes	15.5 / 15.5	2 / 2	1/3/2	0
		C	20	yes	20.3 / 20.3	2 / 2	1/4/1	0
		C'	2.5	yes	2.7 / 2.7	6 / 6	2/4/0	0
		C'	5	yes	4.7 / 5.4	5 / 6	2/3/1	0
		C'	10	yes	10.1 / 10.1	3 / 5	3/1/2	-3
		C'	15	no	15.5 / 16.2	2 / 1	1/3/2	0
		C'	20	yes	20.3 / 20.3	2 / 1	1/3/2	0
	IF s1	C	2.5	yes	2.7 / 2.0	6 / 7	1/3/2	0
		C	5	yes	4.7 / 4.7	6 / 7	0/4/2	0
		C	10	no	9.5 / 8.8	6 / 2	3/3/0	-1
		C	15	yes	14.9 / 14.9	2 / 2	1/4/1	0
		C	20	yes	20.3 / 20.3	2 / 2	1/4/1	0
		C'	2.5	yes	2.7 / 2.0	6 / 6	1/5/0	0
		C'	5	yes	4.7 / 5.4	6 / 6	1/5/0	0
		C'	10	yes	9.5 / 10.1	6 / 6	2/3/1	0
		C'	15	yes	14.9 / 14.9	2 / 2	1/3/2	0
		C'	20	yes	20.3 / 20.3	2 / 1	1/3/2	0
	IF s2	C	2.5	yes	2.0 / 2.7	7 / 7	2/2/2	0
		C	5	yes	5.4 / 4.7	6 / 7	1/3/2	0
		C	10	yes	9.5 / 9.5	5 / 5	2/1/3	0
		C	15	no	15.5 / 14.2	2 / 2	1/4/1	0
		C	20	no	20.3 / 18.9	2 / 2	1/4/1	0
		C'	2.5	yes	2.0 / 2.0	7 / 6	5/1/0	-3
		C'	5	yes	5.4 / 4.7	6 / 6	2/4/0	0
		C'	10	yes	9.5 / 10.1	5 / 6	3/1/2	-1
		C'	15	yes	15.5 / 15.5	2 / 2	1/2/3	0
		C'	20	yes	20.3 / 20.3	2 / 1	1/3/2	0
	LSTM s0	C	2.5	yes	2.7 / 2.7	10 / 5	4/2/0	-8
		C	5	yes	4.7 / 4.7	7 / 5	3/2/1	-2
		C	10	yes	10.1 / 10.1	4 / 2	2/3/1	0
		C	15	yes	14.9 / 14.9	4 / 2	2/2/2	0
		C	20	yes	20.3 / 19.6	2 / 1	1/3/2	0
		C'	2.5	yes	2.7 / 2.0	10 / 5	4/1/1	-3
		C'	5	yes	4.7 / 4.7	7 / 3	5/0/1	-8
		C'	10	yes	10.1 / 9.5	4 / 3	4/1/1	-1
		C'	15	yes	14.9 / 14.9	4 / 1	3/3/0	-1
		C'	20	yes	20.3 / 20.3	2 / 1	3/2/1	-1
	LSTM s1	C	2.5	yes	2.7 / 2.7	8 / 5	4/2/0	-4
		C	5	yes	4.7 / 5.4	7 / 5	4/2/0	-3
		C	10	yes	10.1 / 9.5	1 / 2	1/2/3	0
		C	15	yes	14.9 / 15.5	1 / 1	1/2/3	0
		C	20	yes	20.3 / 19.6	1 / 1	1/3/2	0
		C'	2.5	yes	2.7 / 2.7	8 / 5	5/0/1	-4
		C'	5	yes	4.7 / 5.4	7 / 5	4/0/2	-4
		C'	10	yes	10.1 / 9.5	1 / 3	1/4/1	0
		C'	15	yes	14.9 / 15.5	1 / 1	1/3/2	0
		C'	20	yes	20.3 / 20.3	1 / 1	1/3/2	0
	LSTM s2	C	2.5	yes	2.7 / 2.0	9 / 5	4/1/1	-4
		C	5	yes	4.7 / 4.7	5 / 5	3/2/1	-4
		C	10	yes	10.1 / 9.5	5 / 2	2/2/2	0
		C	15	yes	14.9 / 14.9	5 / 1	2/3/1	0
		C	20	yes	20.3 / 20.3	2 / 1	2/2/2	0
		C'	2.5	yes	2.7 / 2.7	9 / 5	5/0/1	-4
		C'	5	yes	4.7 / 4.7	5 / 3	4/1/1	-4
		C'	10	yes	10.1 / 9.5	5 / 3	2/3/1	0
		C'	15	no	14.9 / 14.2	5 / 1	2/4/0	0
		C'	20	no	20.3 / 18.9	2 / 1	2/2/2	0

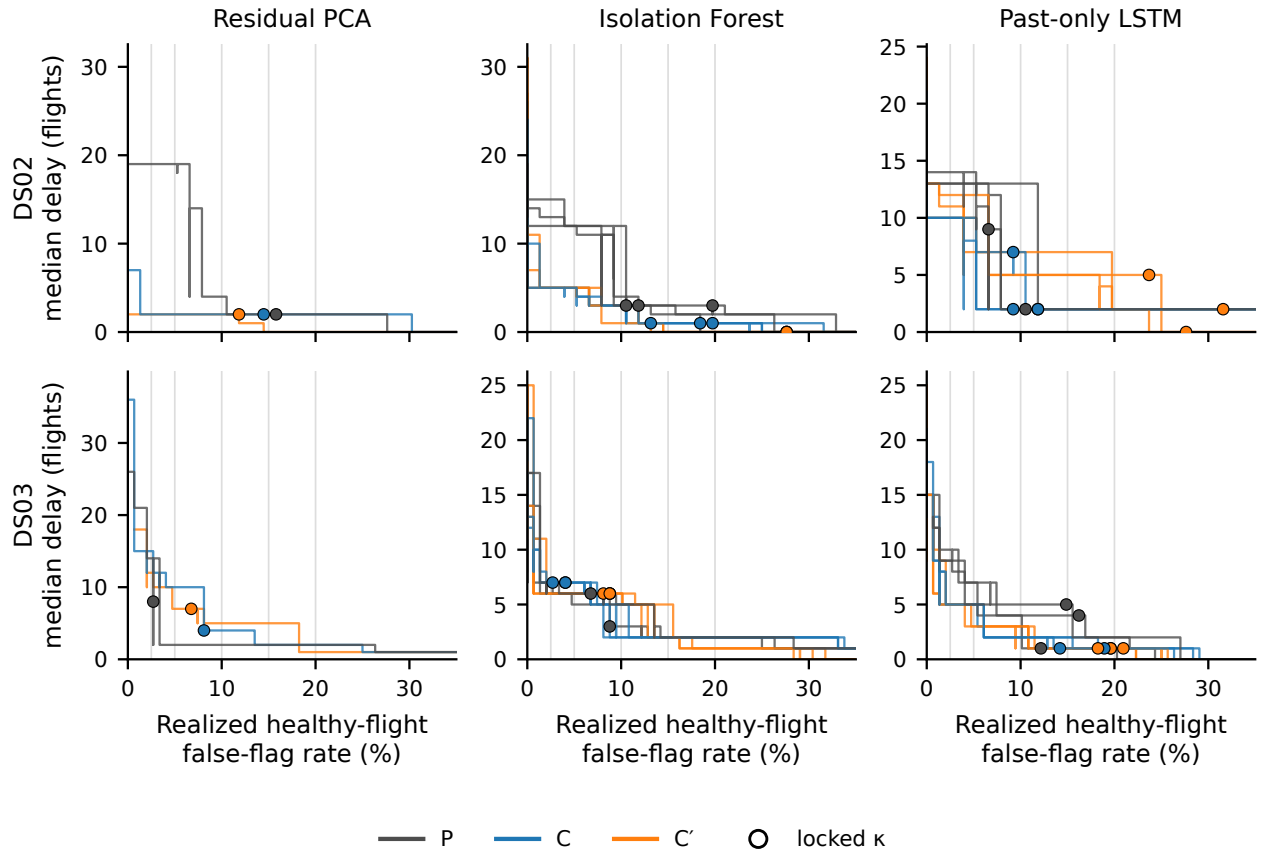


Figure S3: Median detection delay against realized pooled healthy-flight false-flag rate at the 1% row target, one step curve per seed and arm over the calibration-only κ grid. Grey vertical lines mark the matching anchors; filled markers are the locked- κ operating points (nominal 5%). Vertical segments mark κ intervals that share a realized rate.

Table S14: Pre-specified run labels at the 1% target under the nearest-anchor rule and the conservative (not-exceeding) sensitivity rule, curve-level mean delays over realized false-flag rates 2.5–20% (flights), and the Pareto class of the locked operating point against P. Labels: E, earlier at matched burden; L, later; M, mixed; =, equal; n/e, not evaluable.

Dataset	Run	Arm	Supported anchors	Label (nearest)	Label (conservative)	Curve mean P / arm	Locked point
DS02	PCA	C	4	M	M	6.93 / 2.00	arm dominates
	IF s0	C	4	E	E	6.09 / 2.38	trade-off
	IF s1	C	5	E	E	7.17 / 2.38	arm dominates
	IF s2	C	5	E	E	5.75 / 2.38	trade-off
	LSTM s0	C	3	M	M	6.74 / 3.23	identical
	LSTM s1	C	5	M	M	4.87 / 4.91	trade-off
	LSTM s2	C	3	M	M	4.85 / 3.01	trade-off
	PCA	C'	4	E	E	6.93 / 1.52	arm dominates
	IF s0	C'	4	E	E	6.09 / 2.33	trade-off
	IF s1	C'	5	E	E	7.17 / 2.27	trade-off
	IF s2	C'	5	E	E	5.75 / 1.58	trade-off
	LSTM s0	C'	3	M	M	6.74 / 5.97	arm dominated
	LSTM s1	C'	5	M	M	4.87 / 7.41	arm dominated
	LSTM s2	C'	3	M	M	4.85 / 6.09	trade-off
DS03	PCA	C	3	M	E	2.61 / 5.70	trade-off
	IF s0	C	5	M	M	3.54 / 3.28	trade-off
	IF s1	C	4	=	=	4.09 / 3.96	trade-off
	IF s2	C	3	=	=	4.53 / 3.84	trade-off
	LSTM s0	C	5	M	M	4.83 / 2.26	trade-off
	LSTM s1	C	5	M	=	3.00 / 2.31	arm dominated
	LSTM s2	C	5	M	E	4.72 / 2.42	arm dominates
	PCA	C'	3	M	M	2.61 / 5.59	trade-off
	IF s0	C'	4	M	M	3.54 / 3.94	arm dominated
	IF s1	C'	5	=	M	4.09 / 3.96	arm dominates
	IF s2	C'	5	M	E	4.53 / 4.23	arm dominated
	LSTM s0	C'	5	E	E	4.83 / 2.25	trade-off
	LSTM s1	C'	5	M	=	3.00 / 2.24	arm dominated
	LSTM s2	C'	3	E	E	4.72 / 2.12	trade-off

Table S15: Per-engine detection delay (flights from the labelled abnormal-state onset; cens. = no flagged post-onset flight) and healthy-flight false-flag rate at the locked κ (1% row target).

Dataset	Run	Engine	P: delay / FFR (%)	C: delay / FFR (%)	C': delay / FFR (%)
DS02	PCA	11	2 / 5.6	10 / 0.0	10 / 0.0
		14	4 / 17.1	1 / 28.6	1 / 25.7
		15	2 / 21.7	2 / 4.3	2 / 0.0
	IF s0	11	3 / 5.6	3 / 5.6	3 / 5.6
		14	4 / 17.1	1 / 34.3	0 / 54.3
		15	2 / 4.3	0 / 4.3	0 / 4.3
	IF s1	11	3 / 5.6	3 / 5.6	3 / 5.6
		14	4 / 25.7	1 / 22.9	0 / 54.3
		15	2 / 21.7	0 / 4.3	0 / 4.3
	IF s2	11	3 / 5.6	1 / 5.6	3 / 5.6
		14	4 / 17.1	1 / 37.1	0 / 54.3
		15	2 / 8.7	0 / 4.3	0 / 4.3
	LSTM s0	11	11 / 5.6	7 / 0.0	5 / 5.6
		14	0 / 22.9	1 / 22.9	0 / 57.1
		15	2 / 0.0	2 / 4.3	2 / 13.0
	LSTM s1	11	13 / 5.6	10 / 0.0	5 / 5.6
		14	0 / 20.0	1 / 20.0	0 / 45.7
		15	2 / 0.0	7 / 0.0	5 / 4.3
	LSTM s2	11	11 / 5.6	5 / 0.0	5 / 5.6
		14	9 / 8.6	1 / 14.3	0 / 51.4
		15	2 / 4.3	2 / 8.7	0 / 8.7
DS03	PCA	10	23 / 5.9	17 / 11.8	19 / 11.8
		11	8 / 0.0	2 / 0.0	5 / 5.6
		12	15 / 0.0	10 / 13.9	7 / 0.0
		13	2 / 5.6	2 / 11.1	1 / 16.7
		14	14 / 0.0	4 / 0.0	7 / 2.9
	IF s0	15	1 / 8.3	12 / 12.5	11 / 12.5
		10	18 / 5.9	19 / 5.9	18 / 11.8
		11	3 / 11.1	7 / 0.0	6 / 5.6
		12	0 / 8.3	1 / 2.8	0 / 8.3
		13	18 / 11.1	18 / 11.1	18 / 0.0
		14	5 / 11.4	6 / 5.7	2 / 11.4

(continued)

Dataset	Run	Engine	P: delay / FFR (%)	C: delay / FFR (%)	C': delay / FFR (%)
	IF s1	15	1 / 4.2	8 / 0.0	6 / 12.5
		10	19 / 11.8	19 / 5.9	18 / 11.8
		11	6 / 11.1	7 / 0.0	6 / 5.6
		12	0 / 5.6	0 / 2.8	0 / 8.3
		13	18 / 11.1	18 / 11.1	18 / 0.0
	IF s2	14	6 / 11.4	6 / 5.7	3 / 11.4
		15	1 / 4.2	8 / 0.0	6 / 8.3
		10	19 / 5.9	19 / 5.9	18 / 11.8
		11	6 / 5.6	7 / 0.0	6 / 5.6
		12	3 / 5.6	4 / 2.8	0 / 8.3
	LSTM s0	13	18 / 11.1	18 / 11.1	18 / 0.0
		14	6 / 8.6	6 / 0.0	5 / 11.4
		15	6 / 4.2	8 / 0.0	6 / 12.5
		10	6 / 35.3	6 / 41.2	5 / 29.4
		11	5 / 11.1	2 / 22.2	5 / 16.7
	LSTM s1	12	4 / 5.6	1 / 11.1	0 / 19.4
		13	1 / 38.9	1 / 33.3	0 / 27.8
		14	5 / 8.6	5 / 8.6	3 / 11.4
		15	1 / 16.7	1 / 20.8	1 / 20.8
		10	5 / 41.2	6 / 47.1	5 / 35.3
	LSTM s2	11	2 / 11.1	2 / 16.7	5 / 16.7
		12	0 / 0.0	0 / 11.1	0 / 22.2
		13	1 / 27.8	1 / 27.8	1 / 27.8
		14	6 / 0.0	5 / 8.6	3 / 11.4
		15	0 / 16.7	1 / 20.8	1 / 20.8
		10	5 / 41.2	8 / 35.3	5 / 29.4
		11	5 / 22.2	5 / 11.1	5 / 22.2
		12	9 / 2.8	1 / 8.3	0 / 16.7
		13	1 / 33.3	1 / 27.8	1 / 22.2
		14	7 / 0.0	5 / 0.0	5 / 11.4
		15	1 / 16.7	1 / 20.8	1 / 16.7

Table S16: Abnormal-state ($hs = 0$) row alarm rates (%) at the 1% target and the S4 separability of calibration p-values. S4 pAUC is the normalized partial area under the row-level ROC curve of abnormal versus healthy audit rows over row FPR in $[0, 2\%]$, computed on calibration p-values; it is threshold-free and is not a flight-level operating characteristic. Family guardrail ratios (arm/P, all post-onset rows) ranged 0.98–1.16.

Dataset	Run	Arm	All post-onset (%)	First 10 post-onset flights (%)	S4 pAUC, all	S4 pAUC, first 10
DS02	PCA	P	50.78	1.91	0.445	0.015
		C	59.00	4.03	0.517	0.032
		C'	59.00	5.03	0.529	0.039
	IF s0	P	61.00	1.89	0.537	0.017
		C	67.12	13.23	0.564	0.067
		C'	67.48	13.48	0.573	0.074
	IF s1	P	59.96	1.64	0.519	0.014
		C	66.44	11.44	0.565	0.067
		C'	67.42	12.75	0.569	0.071
	IF s2	P	62.05	2.06	0.545	0.018
		C	68.43	15.50	0.582	0.081
		C'	69.05	16.12	0.585	0.085
	LSTM s0	P	62.41	1.52	0.569	0.016
		C	63.62	4.18	0.583	0.034
		C'	65.98	5.17	0.572	0.028
	LSTM s1	P	63.66	1.43	0.575	0.015
		C	64.62	3.07	0.589	0.030
		C'	66.63	3.76	0.578	0.025
	LSTM s2	P	63.43	1.99	0.574	0.020
		C	63.79	4.19	0.581	0.034
		C'	65.25	5.08	0.565	0.028
DS03	PCA	P	39.51	1.37	0.331	0.010
		C	41.98	1.54	0.337	0.010
		C'	42.53	1.59	0.351	0.010
	IF s0	P	58.00	2.15	0.494	0.015
		C	58.57	2.21	0.502	0.015
		C'	56.87	2.37	0.505	0.018
	IF s1	P	56.88	1.86	0.482	0.014
		C	57.47	1.94	0.492	0.014
		C'	55.79	1.93	0.495	0.016
	IF s2	P	56.87	1.92	0.480	0.014
		C	57.50	2.10	0.490	0.014

(continued)

Dataset	Run	Arm	All post-onset (%)	First 10 post-onset flights (%)	S4 pAUC, all	S4 pAUC, first 10
LSTM s0		C'	56.23	2.19	0.493	0.017
		P	58.37	2.11	0.497	0.012
		C	58.15	2.62	0.499	0.015
LSTM s1		C'	57.64	3.76	0.500	0.018
		P	58.13	2.33	0.489	0.013
		C	57.87	2.71	0.492	0.015
LSTM s2		C'	57.26	3.95	0.496	0.019
		P	59.26	2.29	0.507	0.012
		C	59.05	2.65	0.508	0.015
		C'	58.85	4.70	0.512	0.021

Table S17: Healthy flights with any alarm (the frozen burden endpoint) under P, C and C' at the 1% target, shown beside the κ -based healthy-flight false-flag rate. The two quantities are distinct and are never merged.

Dataset	Run	Arm	Any alarm, overall (%)	Any alarm, climb (%)	Any alarm, cruise (%)	Any alarm, descent (%)	κ -based FFR (%)
DS02	PCA	P	98.7	13.2	50.0	98.7	15.8
		C	100.0	56.6	81.6	86.8	14.5
		C'	96.1	44.7	81.6	68.4	11.8
	IF s0	P	100.0	68.4	44.7	86.8	10.5
		C	100.0	97.4	81.6	59.2	18.4
		C'	100.0	96.1	73.7	65.8	27.6
	IF s1	P	98.7	60.5	44.7	85.5	19.7
		C	100.0	97.4	77.6	64.5	13.2
		C'	100.0	96.1	72.4	68.4	27.6
	IF s2	P	100.0	67.1	46.1	86.8	11.8
		C	100.0	97.4	82.9	56.6	19.7
		C'	100.0	97.4	76.3	67.1	27.6
	LSTM s0	P	100.0	97.4	77.6	100.0	11.8
		C	100.0	100.0	90.8	88.2	11.8
		C'	100.0	100.0	90.8	71.1	31.6
	LSTM s1	P	100.0	92.1	64.5	100.0	10.5
		C	100.0	100.0	85.5	69.7	9.2
		C'	100.0	100.0	90.8	67.1	23.7
	LSTM s2	P	100.0	98.7	68.4	100.0	6.6
		C	100.0	100.0	78.9	76.3	9.2
		C'	100.0	100.0	90.8	65.8	27.6
DS03	PCA	P	100.0	67.6	73.6	99.3	2.7
		C	100.0	91.9	84.5	99.3	8.1
		C'	100.0	92.6	87.2	99.3	6.8
	IF s0	P	100.0	81.1	67.6	99.3	8.8
		C	100.0	99.3	69.6	99.3	4.1
		C'	100.0	93.2	64.9	99.3	8.8
	IF s1	P	100.0	81.1	65.5	99.3	8.8
		C	100.0	99.3	68.2	98.6	4.1
		C'	100.0	93.9	63.5	99.3	8.1
	IF s2	P	100.0	77.7	66.2	99.3	6.8
		C	100.0	98.0	68.2	99.3	2.7
		C'	100.0	93.9	63.5	98.0	8.8
	LSTM s0	P	100.0	100.0	95.3	100.0	16.2
		C	100.0	100.0	95.3	99.3	19.6
		C'	100.0	100.0	90.5	89.2	19.6
	LSTM s1	P	100.0	100.0	92.6	100.0	12.2
		C	100.0	100.0	91.9	99.3	18.9
		C'	100.0	100.0	92.6	87.8	20.9
	LSTM s2	P	100.0	100.0	94.6	99.3	14.9
		C	100.0	100.0	92.6	98.0	14.2
		C'	100.0	100.0	89.9	87.2	18.2

Continuous contextual baseline

Table S18: Continuous contextual baseline Q: linear quantile regression of the detector score at level $1 - \alpha$ on a quadratic surface of the four standardized operating descriptors (4 linear, 4 squared and 6 pairwise terms plus an intercept), solved exactly by linear programming (HiGHS, no penalty) on every fifth healthy calibration row, per run and target, with no tuning and no phase input. Fitted coefficients are in the machine-readable evidence files.

Dataset	Target	Run	Fit rows	In-sample alarm rate (%)	κ	Audit FFR at κ (%)
DS02	0.5%	PCA	77,329	0.52	0.0195	21.1
		IF s0	77,329	0.53	0.0116	31.6
		IF s1	77,329	0.52	0.0116	31.6
		IF s2	77,329	0.52	0.0106	32.9
		LSTM s0	77,329	0.52	0.0137	19.7
		LSTM s1	77,329	0.52	0.0164	18.4
		LSTM s2	77,329	0.52	0.0201	17.1
	1%	PCA	77,329	1.03	0.0434	9.2
		IF s0	77,329	1.00	0.0307	18.4
		IF s1	77,329	1.01	0.0235	17.1
		IF s2	77,329	1.00	0.0318	18.4
		LSTM s0	77,329	0.98	0.0247	18.4
		LSTM s1	77,329	1.03	0.0304	7.9
		LSTM s2	77,329	1.01	0.0250	22.4
	2%	PCA	77,329	1.99	0.0618	11.8
		IF s0	77,329	1.98	0.0704	17.1
		IF s1	77,329	2.00	0.0651	13.2
		IF s2	77,329	2.01	0.0864	9.2
		LSTM s0	77,329	1.99	0.0571	6.6
		LSTM s1	77,329	2.04	0.0647	6.6
		LSTM s2	77,329	2.03	0.0535	9.2
DS03	0.5%	PCA	91,621	0.49	0.0169	7.4
		IF s0	91,621	0.50	0.0149	11.5
		IF s1	91,621	0.50	0.0143	11.5
		IF s2	91,621	0.48	0.0149	10.1
		LSTM s0	91,621	0.50	0.0158	5.4
		LSTM s1	91,621	0.51	0.0173	3.4
		LSTM s2	91,621	0.48	0.0167	3.4
	1%	PCA	91,621	1.00	0.0394	6.1
		IF s0	91,621	1.00	0.0359	10.8
		IF s1	91,621	1.00	0.0448	5.4
		IF s2	91,621	1.00	0.0449	5.4
		LSTM s0	91,621	1.00	0.0305	7.4
		LSTM s1	91,621	0.97	0.0292	7.4
		LSTM s2	91,621	0.99	0.0340	4.1
	2%	PCA	91,621	2.05	0.1129	2.0
		IF s0	91,621	2.02	0.1079	5.4
		IF s1	91,621	2.02	0.1170	4.1
		IF s2	91,621	2.02	0.1249	3.4
		LSTM s0	91,621	2.01	0.0547	10.8
		LSTM s1	91,621	2.02	0.0613	9.5
		LSTM s2	91,621	2.03	0.0634	7.4

Table S19: Continuous contextual baseline Q: pooled audit healthy FPR (%) by phase and calibration metrics at the 1% target (healthy side only).

Dataset	Run	Climb	Cruise	Descent	Overall	A1 (pp)	A2 (pp)	A3 (pp)
DS02	PCA	0.702	2.125	1.271	1.462	1.42	1.12	0.69
	IF s0	0.490	2.239	2.138	1.742	1.75	1.24	1.01
	IF s1	0.421	1.339	1.937	1.296	1.52	0.94	0.67
	IF s2	0.563	2.455	2.113	1.840	1.89	1.46	1.09
	LSTM s0	0.717	1.546	1.294	1.242	0.83	0.55	0.39
	LSTM s1	0.562	1.289	0.967	0.989	0.73	0.44	0.30
	LSTM s2	0.819	2.711	3.140	2.353	2.32	2.14	1.59
DS03	PCA	2.657	1.798	1.405	1.853	1.25	1.66	1.09
	IF s0	1.959	1.607	1.704	1.735	0.35	0.96	0.77
	IF s1	1.869	1.502	1.509	1.597	0.37	0.87	0.65
	IF s2	1.903	1.461	1.589	1.625	0.44	0.90	0.68
	LSTM s0	1.513	1.121	1.690	1.454	0.57	0.69	0.50
	LSTM s1	1.724	1.192	1.714	1.541	0.53	0.72	0.60
	LSTM s2	1.493	1.064	1.630	1.405	0.57	0.63	0.46

Operating support and the phase effect within calibration engines

Table S20: Operating-support distances of healthy audit rows to the nearest healthy calibration row of the same retrospective phase (standardized altitude, Mach number, throttle-resolver angle and fan-inlet temperature), with the any-phase and 8-D (adding 120-s trailing slopes) variants, the fraction beyond the cross-calibration-engine 95th-percentile reference, and the fraction outside the calibration range.

Dataset	Engine (class)	Phase	Rows	Median d	95th pct d	Median d , any phase	Median d , 8-D	Beyond ref. (%)	Outside range (%)
DS02	11 (3)	climb	53,678	0.024	0.137	0.022	0.131	1.2	0.2
		cruise	86,535	0.024	0.348	0.013	0.152	0.0	0.0
		descent	73,093	0.050	0.256	0.043	0.292	2.6	0.0
	14 (1)	climb	12,241	0.045	0.280	0.042	0.240	4.6	2.3
		cruise	33,377	1.130	1.443	0.089	1.324	38.6	95.3
		descent	26,650	0.060	0.216	0.060	0.255	0.3	22.4
	15 (2)	climb	48,789	0.037	0.126	0.035	0.168	1.8	0.6
		cruise	53,466	0.068	0.319	0.023	0.241	0.0	1.4
		descent	45,284	0.049	0.193	0.048	0.306	0.2	0.1
DS03	10 (3)	climb	63,828	0.039	0.248	0.032	0.153	1.8	3.6
		cruise	95,181	0.069	0.428	0.048	0.178	2.4	7.3
		descent	90,839	0.043	0.238	0.037	0.251	0.5	0.6
	11 (3)	climb	63,479	0.026	0.138	0.023	0.132	0.0	0.0
		cruise	98,352	0.025	0.329	0.022	0.151	0.0	2.5
		descent	99,376	0.043	0.259	0.037	0.232	1.3	0.1
	12 (1)	climb	35,417	0.022	0.119	0.022	0.175	0.0	0.0
		cruise	47,845	0.736	1.096	0.060	1.036	14.6	84.3
		descent	72,178	0.034	0.149	0.034	0.269	0.0	0.0
	13 (3)	climb	66,056	0.026	0.188	0.025	0.129	0.1	0.5
		cruise	81,213	0.020	0.196	0.018	0.132	0.1	3.0
		descent	104,404	0.038	0.331	0.037	0.224	0.3	4.2
	14 (1)	climb	33,331	0.022	0.102	0.022	0.179	0.0	0.0
		cruise	52,432	0.878	1.108	0.061	1.125	18.0	82.1
		descent	73,818	0.029	0.100	0.029	0.244	0.0	0.0
	15 (2)	climb	64,502	0.020	0.125	0.019	0.117	0.0	0.2
		cruise	61,379	0.025	0.205	0.015	0.125	0.0	1.1
		descent	91,049	0.025	0.142	0.024	0.200	0.0	0.3

Table S21: Spearman correlations across engine $\times$ phase cells between median same-phase support distance and calibration error, per run at the 1% target, with leave-one-engine-out ranges. Pre-specified verdict: indeterminate due to small engine count.

Dataset	Run	Cells	$\rho(d, \text{err} \text{ C})$	$\rho(d, \text{err} \text{ C}')$	$\rho(d_{\text{all}}, \text{err} \text{ P})$	$\rho(d, \text{change } \text{P} \rightarrow \text{C})$	Leave-one-engine-out range
DS02	PCA	9	-0.23	0.33	0.25	-0.05	[-0.60, -0.03]
DS03	PCA	18	0.02	0.03	0.16	-0.11	[-0.25, 0.17]
DS02	IF s0	9	0.37	0.43	0.15	0.10	[0.03, 0.43]
DS03	IF s0	18	0.14	0.08	0.27	-0.11	[-0.10, 0.47]
DS02	IF s1	9	0.03	0.43	0.22	-0.08	[-0.49, 0.20]
DS03	IF s1	18	0.14	0.13	0.19	-0.15	[-0.07, 0.39]
DS02	IF s2	9	0.25	0.43	0.22	0.08	[-0.20, 0.43]
DS03	IF s2	18	0.08	0.08	0.17	-0.17	[-0.11, 0.36]
DS02	LSTM s0	9	-0.22	0.17	0.10	-0.05	[-0.60, -0.03]
DS03	LSTM s0	18	-0.06	-0.02	0.15	-0.40	[-0.35, 0.18]
DS02	LSTM s1	9	0.25	0.30	0.08	0.08	[-0.03, 0.26]
DS03	LSTM s1	18	0.01	-0.04	0.14	-0.26	[-0.31, 0.29]
DS02	LSTM s2	9	-0.17	0.13	0.10	0.08	[-0.54, -0.14]
DS03	LSTM s2	18	0.04	0.02	0.25	-0.24	[-0.23, 0.30]

Table S22: C0: pooled-threshold healthy FPR (%) by phase within the calibration engines at the 1% target, in-sample and from one calibration engine to the other, where calibration-to-audit mismatch is absent by construction.

Dataset	Run	Evaluation	Climb	Cruise	Descent	Overall	Highest
DS02	PCA	in-sample	0.044	0.470	2.416	1.000	descent
		engine 18 to 20	0.015	0.586	2.309	1.057	descent
		engine 20 to 18	0.067	0.371	2.317	0.874	descent
DS03	PCA	in-sample	0.448	0.493	1.742	1.000	descent
		engine 4 to 8	0.176	0.600	1.732	0.911	descent
		engine 8 to 4	0.803	0.250	1.825	1.137	descent
DS02	IF s0	in-sample	0.155	0.383	2.421	1.000	descent
		engine 18 to 20	0.230	0.494	2.367	1.105	descent
		engine 20 to 18	0.094	0.272	2.271	0.828	descent
DS03	IF s0	in-sample	0.356	0.746	1.617	1.000	descent
		engine 4 to 8	0.356	1.113	2.322	1.362	descent
		engine 8 to 4	0.261	0.388	1.102	0.683	descent
DS02	IF s1	in-sample	0.137	0.370	2.450	1.000	descent
		engine 18 to 20	0.215	0.515	2.592	1.191	descent
		engine 20 to 18	0.075	0.249	2.203	0.792	descent
DS03	IF s1	in-sample	0.347	0.732	1.633	1.000	descent
		engine 4 to 8	0.338	1.130	2.328	1.365	descent
		engine 8 to 4	0.264	0.359	1.174	0.710	descent
DS02	IF s2	in-sample	0.143	0.388	2.425	1.000	descent
		engine 18 to 20	0.206	0.486	2.484	1.138	descent
		engine 20 to 18	0.092	0.293	2.302	0.845	descent
DS03	IF s2	in-sample	0.411	0.804	1.537	1.000	descent
		engine 4 to 8	0.360	1.228	2.219	1.364	descent
		engine 8 to 4	0.390	0.376	1.095	0.713	descent
DS02	LSTM s0	in-sample	0.267	0.719	1.943	1.000	descent
		engine 18 to 20	0.347	1.230	2.155	1.322	descent
		engine 20 to 18	0.233	0.398	1.929	0.815	descent
DS03	LSTM s0	in-sample	0.438	0.971	1.397	1.000	descent
		engine 4 to 8	0.570	1.958	2.146	1.650	descent
		engine 8 to 4	0.310	0.420	0.876	0.600	descent
DS02	LSTM s1	in-sample	0.298	0.636	2.012	1.000	descent
		engine 18 to 20	0.284	0.897	1.988	1.125	descent
		engine 20 to 18	0.325	0.462	2.050	0.904	descent
DS03	LSTM s1	in-sample	0.430	1.102	1.306	1.000	descent
		engine 4 to 8	0.592	1.984	2.127	1.658	descent
		engine 8 to 4	0.278	0.434	0.786	0.553	descent
DS02	LSTM s2	in-sample	0.324	0.696	1.921	1.000	descent
		engine 18 to 20	0.359	1.195	2.116	1.299	descent
		engine 20 to 18	0.305	0.427	1.936	0.850	descent
DS03	LSTM s2	in-sample	0.432	1.154	1.266	1.000	descent
		engine 4 to 8	0.593	2.259	2.223	1.790	cruise
		engine 8 to 4	0.288	0.436	0.698	0.515	descent

Post-plan error decomposition

Table S23: Post-plan descriptive decomposition (deviation Dev-1) of the equal-cell mean squared nominal error over audit engine $\times$ phase cells at the 1% target into overall bias, phase, engine and interaction shares.

Dataset	Run	Arm	RMS (pp)	Bias ² (%)	Phase (%)	Engine (%)	Engine $\times$ phase (%)
DS02	PCA	P	1.80	7	85	4	5
		C	1.54	19	13	35	34
		C'	1.96	18	10	45	27
	IF s0	P	0.96	5	72	17	6
		C	2.55	20	14	39	27
		C'	2.38	21	6	53	19
	IF s1	P	1.20	8	76	15	1
		C	2.25	17	14	39	30
		C'	2.26	21	5	54	19
	IF s2	P	1.04	7	74	16	4
		C	2.53	19	15	38	28

(continued)

Dataset	Run	Arm	RMS (pp)	Bias ² (%)	Phase (%)	Engine (%)	Engine×phase (%)
DS03	LSTM s0	C'	2.42	22	7	52	20
		P	0.79	7	51	24	18
		C	1.53	15	15	36	34
	LSTM s1	C'	2.00	31	1	54	14
		P	0.80	8	45	20	27
		C	1.37	15	13	40	33
	LSTM s2	C'	1.75	26	1	55	18
		P	0.82	11	35	23	31
		C	1.61	13	18	34	35
	PCA	C'	1.99	30	2	53	15
		P	0.66	10	57	19	14
		C	0.80	48	12	21	19
	IF s0	C'	0.84	41	18	22	19
		P	0.83	6	70	9	15
		C	0.77	26	5	36	33
	IF s1	C'	0.58	10	37	20	33
		P	0.83	4	68	12	17
		C	0.71	28	6	37	28
	IF s2	C'	0.57	8	39	19	34
		P	0.84	6	65	12	17
		C	0.81	27	6	38	30
	LSTM s0	C'	0.59	14	34	19	33
		P	1.03	25	29	27	20
		C	0.95	45	4	21	31
	LSTM s1	C'	1.01	58	1	16	24
		P	1.09	23	28	29	20
		C	0.98	43	5	23	30
	LSTM s2	C'	1.05	55	3	17	25
		P	1.17	21	22	34	22
		C	0.99	38	4	26	33
		C'	1.11	56	1	16	27

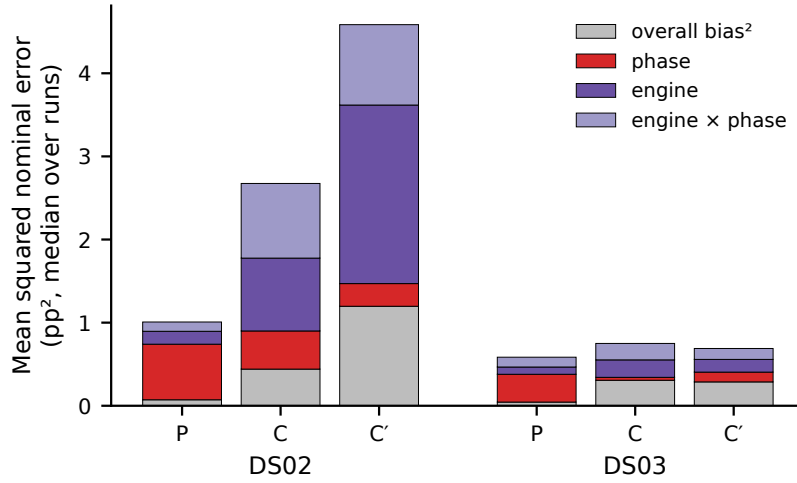


Figure S4: Post-plan descriptive decomposition (Dev-1) of the equal-cell mean squared nominal error at the 1% target (median over the seven detector runs).